

Pinball Machine

Functional Specification

Description: In this game, users manipulate electromechanical flippers to push a ball up an inclined plane, attempting to hit bumper targets to earn points, while keeping the ball from going down the drain behind the flippers. Each game may consist of one or two players. In each game, each player receives 5 balls, or 5 opportunities to score points, alternating turns after each ball. A new game is started by pressing the “Start” button on the front of the machine. Pressing the “Start” button after a game has been started will toggle between one and two players. More games are added to the machine by pressing the “Game” Button.

The Pinball Machine comes with a number of sensors and LEDs along with a speaker for generating sounds and a serial EEROM for storing high scores and number of games left to play. The sensors are located throughout the playing field and when activated cause the score to increase by a determined amount. There are two 4-bit 7-segment displays for indicating the current score. Additionally there are a number of individual lights on the playing field and backbox.

Inputs: Input to the pinball machine comes through an array of 40 switches, 31 of which will be used. All buttons will be debounced. The following table describes each of their functionality

Switch/Input Name	Description
Game Button (Quarter slot)	Activated by the user. Adds 1 to the number of games on the machine, saturating at a maximum of 255. If a game is not currently in play, it will update the display, reading “PLAY” and the number of games left on the machine.
Start Button (red circle, front of machine)	If there are available games on the machine to be played (games_left >= 1), a new game will be started, initializing all player’s scores to 0. The ball will be loaded into the launcher and the Player 1 7-segment display will read “0000” indicating a one player game has been started, with scores reset. Pressing the Start button after the initial press will toggle between a one and two player game. When a two player game is selected, both the Player 1 and Player 2 displays will read “0000”. Each game consists of 5 rounds, or 5 balls, per player, in which they will launch the ball and use the flippers to 1) keep the ball from going down the drain and 2) hit targets on the playing field to score points and other bonuses. When a new game is initialized by hitting the Start Button, three LEDs will be turned on, one for a 200 bonus point light (initially located above the top left rollover button), and one each for the

	Rotating Bumper's point multiplier of the Free Ball targets (see their descriptions for more information). Immediately upon pressing the Start Button, game music will be played from the machine to the tune of "Entry of the Gladiators" by Julius Fucik, more commonly known as circus music. This music will continue until the end of the game, when all 5 balls have been played by all players. If there are not available games on the machine, the Start Button does nothing. If a game is already in play (once a sensor has been hit by the ball after the Start Button pressed) the user can no longer toggle between one and two players, and the Start Button does nothing.
Switch Bonus Leaf Bumpers (x4)	Four "Switch Bonus" Leaf Bumpers are located throughout the board. Two at the very top (white bands diagonal) above the Top Rollover Bumpers, and two on the sides of the playing field, about midway up (also straight white bands). When hit, these bumpers will give the current player 1 point. Additionally, these bumpers will toggle the Bonus Lights from left to right or back. These Bonus Lights affect the point value of the Top Rollover Bumpers (see below) and which of the Free Ball Target's point value is incremented when the Rotating Bumper is hit.
Top Rollover Bumpers	Two Rollover Bumpers are located at the top of the board, which the ball must hit in order to go over to get to the main playing field. Each of these bumpers give the current player 100 points when rolled over. However, the rollover bumper corresponding to that of the Bonus Light will give an additional 200 points (for 300 total). So if the Bonus Light is above the Left Rollover Bumper, then the Left Rollover Bumper will give 300 points instead of 100. Same for the Right.
Card Targets (x8)	8 leaf bumpers located towards the top of the pinball machine under the depicted playing cards. Four cards, on the left, are Aces, and the four on the right are Kings. When a card is hit, an LED will be turned on under that card, and a player will receive 10 points. The LEDs that are lit remain on until they are reset by their corresponding Free Ball Target, or when the ball goes down the drain. Hitting all four of one card type (i.e. all four of the Ace card targets on the left, or all King targets on the right) will make a free ball award available on a corresponding Free Ball target. This will be displayed by an LED near the Free Ball target being lit. If the ball goes into the free ball target while that LED is lit, they will receive an extra ball that round. The King cards, on the right, correspond to the Right Free Ball Target. The Ace cards on

	the left correspond to the Left Free Ball Target.
Rotating Bumper	A red spinner is located in the center on the board. This target gives no points but increases the point value for one of the Free Ball Targets. Which of the two Free Ball targets receives the point increase is dependent on the Bonus Lights located near the bumper. If the left light is lit, then rotating the bumper increases the point value of the Left Free Ball target by 10 points, up to 100. Likewise if the right light is lit when the rotating bumper is hit, the point value of the Right Free Ball Target is increased by 10 points, up to 100. Two rows of 10 LEDs are on either side of the Rotating Bumper. Each time the Rotating Bumper is hit, an LED will be lit in a row, and next to a label with the new point value, of the corresponding Free Ball Target. These lights and the extra points are reset to one light, and 10 points, when the relevant Free Ball Target is hit.
Free Ball Targets (x2)	Two Free Ball targets are located on the playing field, one on the left, and one on the right. When a Free Ball Target is hit, the current player will receive the number of points indicated by the rows of LEDs next to the Rotating Bumper (Left row for the Left Target, right row for Right target). If one light is lit, the player will receive 10 points, 2 lights are 20 points, etc, up to 100 points for all 10 lights being lit. Point values are indicated on the board. Hitting a Free Ball Target will turn off all but the first light and reset the point value of that Target to 10 points. The other Free Ball Target will not be affected. If the Free Ball available light is lit next to the Free Ball Target that the ball rolls into, the player will receive a free ball at the end of their turn. A light saying “Same Player Go Again” will be lit, telling the user that the same player will go again directly after their ball goes down the drain. When the ball goes down the drain, it will immediately be loaded into the launcher for the same player, and the “Same Player Go Again” light will go off. Additionally, hitting a Free Ball Target that has a free ball available will reset the Free Ball available light, and also reset the lights of the Kings or Aces card buttons corresponding to the Free Ball Target (Kings for right, Aces for left).
Pop Bumpers (x3)	When activated by the ball, a pop bumper will cause a corresponding LED light to flash for a quarter second (250 milliseconds), and give the current player 10 points. Three pop bumpers, shaped like mushrooms, are in the top middle of the playing field.
‘Generic’ Bounce	Four generic bounce leaf bumpers are located throughout the

Leaf Bumpers (x4)	playing field, two on the bottom, one on the outside side of each of the flippers, and two above, one on each side of the board. When hit, these bumpers will flash a light for $\frac{1}{4}$ second (250 ms), and give the current player 1 point.
Bottom Rollover Bumpers	Two Rollover Bumpers are located on the bottom left and right sides of the playing field next to the walls, before the ball would go down the drain. Each of these gives 100 points to the current player when pressed.
Drain	At the bottom of the playing field, past the flippers, is the drain, where the ball goes to die (or end the current round). If there are any balls left in the game (after the ball/round that just finished), then the drain button will blink the display several times with the Player 1 display reading “BALL” and the Player 2 display reading the number of balls the current player has left. Then the ball is loaded into the launcher. When “BALL” is done blinking, the display will be empty until the ball hits a sensor (oh the suspense!). In a two player game, this flashing display marks the time to switch turns (from Player 1 to 2 or the other way around). If there are no balls left in the game (Player 1 used all 5 balls, or player 1 and player 2 used all of the balls, 5 each, for a two player game), then the game will end, and the ball will not be loaded into the launcher. This is the end of the current game. At this point, the game music (circus music) will be stopped, and a low tone will be played through the speaker for a short time. The display will also blink, reading “HIGH” on the Player 1 display, and the high score on the Player 2 display, so the users can see if they beat the high score. If the current player received a Free Ball in the last round then the “Same Player Go Again” light will be lit. In this case, when the ball goes down the drain, the ball will be loaded and the display will NOT blink, indicating that the same player should go again (this is their free ball). The “Same Player Go Again” light will then be turned off. The drain will also turn off all lights turned on due to the Rotating Bumper and the Card Targets, and reset the point values for the Free Ball Targets.
Tilt	There is an electromechanical sensor that detects whether the game board is being tilted or shaken too much. When activated, the current player will lose their ball, and they will not be able to score points until the next round. Lights may still be turned on/off.

Outputs:

4-digit LED Displays:

Two 4-digit LED displays are used for the following purposes: When a game is not in play, it will display “PLAY” on the Player 1 display and the current number of games on the machine on the Player 2 display. During play, the Player 1 display will display Player 1’s points, and the Player 2 display will show Player 2’s points. In a one player game, the Player 2 display will just read zero.

When selecting the number of players in a game, the Start Button will toggle between just the Player 1 display showing all zeroes (one player) and both displays showing all zeroes (two player). The final selection is made once the player hits a target on the board.

At the end of a round, when the ball goes down the drain, the display will blink on and off several times reading “BALL” on the Player 1 display and the number of balls the current player (who just finished their turn) has left on the Player 2 display.

At the end of a game, the Player 1 Display will blink and read “HIGH” while the Player 2 display reads the high score on the machine and blinks. Then go back to the not in play display above.

LEDs:

64 LEDs flash or turn on during game play as different buttons are hit by the ball.

Two rows of 10 LEDs (20 total) will be controlled by the Rotating Bumper. Each time it is rotated another light in the row corresponding to which of the Bonus lights is lit. (e.g. One more on the left lit when the bonus light on the left is lit). All but the first are turned off in each row when the corresponding Free Ball Target is hit, or in both rows when the drain button is pressed.

Two rows of 4 LEDs (8 total) will be controlled by the Ace and King Card Targets, turning on when the corresponding Card Target is pressed. These will remain on until the Left or Right Free Ball Target is activated, respectively, or the drain button is pressed.

Four LEDs are used to show which side of the board receives the bonus points (For the Rotating Bumper/Free Ball Targets and Top Rollover Bumpers). Two on the left are initially lit, and these are both toggled together back and forth between the left and the right each time a Switch Bonus Bumper is hit.

4 LEDs correspond to generic bounce Leaf Bumpers and one is flashed for ¼ second (250 ms) each time its corresponding leaf bumper is hit.

Speaker:

While a game is in play (from when the Start Button is pressed until the last ball goes down the drain) the speaker plays circus music (from “Entry of the Gladiators” by Julius Fucik) on repeat. This is ended by the ending of a game.

The speaker will also play a low tone for a few seconds when a game ends after turning off the music.

Loader:

This mechanism moves the ball from the drain to the launcher. This occurs at the start of a new game or automatically during a game when there are more than 0 balls left after the Drain Button is pressed.

User Interface: The UI consists of the “Game” button (front of the machine), the “Start” button, the electromechanical flipper buttons (side of the machine), and the mechanical launcher. The User presses the “Game” button once to add one game (it may be pressed many times to increment **games** up to 255). Whenever the player presses the “Start” button, **games** is decremented, balls for each player is set to 5, the score is reset to zero, and the loader loads the ball into the launcher. Players then take turns according to LEDs which correspond to whose turn it is. Players launch the ball and use the flippers to keep the ball from going into the drain and hit targets to score points. The speaker makes noises, lights flash, and the score is updated (on two four-digit displays) during play and as the ball hits various targets on the board.

Error Handling: Power Failure:

If there is a power failure, when power comes back on the 4-digit LED Displays will have the default display, showing the word “PLAY” and the number of games left on the machine. Any game that was in play, and its score, will be lost. The number of games left on the machine and the high score will not be affected by a power failure.

Tilt:

If the machine is shaken or tilted too much (measured mechanically and sent electronically to a switch) the score will stop incrementing until the ball goes down the drain, at which point the loader will act according to the number of balls left in the game. The Drain Button will still activate, resetting the lights and decrementing the number of balls the player has.

Free Ball Points, etc:

The maximum points per Free Ball Target (per hit) is 100. If the Rotating Bumper is rotated more times the point values do not change until the corresponding Free Ball Target is hit.

Points:

If a player receives more than 9999 points, the score will wrap to zero.

Algorithms: No advanced algorithms are used. The pinball machine runs off of addition to calculate points and comparisons to set the High Score, point values, and other outputs. Look up tables may be used for inputs and outputs. Input switches are debounced.

Global Variables: None.

Data Structures: Lookup tables are used to store sound music constants, light displays, and function calls with arguments.

Limitations: Limitations dealing with Software:

The maximum number of games that may be on the machine is 255.

The maximum possible score per player is 9999.

The maximum points for a Free Ball Target is 100.

Limitations dealing with Hardware:

Lights can be only on, off, or flash on for 250 milliseconds and then turn off.
Mechanical failures cannot be detected nor fixed automatically.

Known Bugs: No known bugs.

Special Notes: Have fun!